

TP 1 Transversal-EDA “Rendimiento Estudiantil”

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d. Importación y descripción del dataset

```
library(tidyverse)
df <- read_delim("dataset.csv", delim = ";", show_col_types = FALSE)
head(df)
```

```
# A tibble: 6 x 11
  student_id gender    age study_hours_per_week attendance_rate parent_education
  <chr>      <chr>  <dbl>          <dbl>          <dbl> <chr>
1 STU0001   Male     15              25             63.8 Bachelor
2 STU0002   Female   15              2             54.7 Bachelor
3 STU0003   Female   19             10             90.5 High School
4 STU0004   Male     16             26             66.8 High School
5 STU0005   Female   15             25             73   High School
6 STU0006   Female   19              8             85.2 High School
# i 5 more variables: internet_access <chr>, extracurricular <chr>,
#   previous_score <dbl>, final_score <dbl>, passed <chr>
```

```
dim(df)
```

```
[1] 500  11
```

El dataset contiene 500 estudiantes y 11 variables que describen su perfil académico y personal. Fue descargado automáticamente mediante el script descarga.sh disponible en el repositorio de GitHub. 1. Identificación de variables `glimpse(df)`

2. Variables categóricas:

student_id: identificador del estudiante

gender: género (Male/Female)

parent_education: nivel educativo del padre

internet_access: acceso a internet (Yes/No)

extracurricular: actividades extraescolares (Yes/No)

passed: si aprobó o no (Yes/No)

Variables numéricas:

age: edad

study_hours_per_week: horas de estudio semanales

attendance_rate: porcentaje de asistencia

previous_score: nota del período anterior

final_score: nota final

3. Medidas de posición

3.1 Medidas de tendencia central

```
::: {.cell}
```

```
```{r .cell-code}
```

```
vars_num <- df %>% select(age, study_hours_per_week, attendance_rate, previous_score, final_score)
```

```
print("MEDIA:")
```

```
[1] "MEDIA:"
```

```
vars_num %>% summarise(across(everything(), mean))
```

```
A tibble: 1 x 5
```

```
 age study_hours_per_week attendance_rate previous_score final_score
```

	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	17.0	15.3	76.4	63.0	56.0

```
print("MEDIANA:")
```

```
[1] "MEDIANA:"
```

```
vars_num %>% summarise(across(everything(), median))
```

```
A tibble: 1 x 5
 age study_hours_per_week attendance_rate previous_score final_score
 <dbl> <dbl> <dbl> <dbl> <dbl>
1 17 15 76.5 64 56
```

```
:::
```

## Moda (valor más frecuente)

```
moda <- function(x) as.numeric(names(sort(table(x), decreasing = TRUE)[1]))
print("MODA:")
```

```
[1] "MODA:"
```

```
vars_num %>% summarise(across(everything(), moda))
```

```
A tibble: 1 x 5
 age study_hours_per_week attendance_rate previous_score final_score
 <dbl> <dbl> <dbl> <dbl> <dbl>
1 15 10 72.7 68 45
```

## 3.2 Cuartiles

```
vars_num %>%
 summarise(across(everything(), list(
 Q1 = ~ quantile(.x, 0.25),
 Q2 = ~ quantile(.x, 0.50),
 Q3 = ~ quantile(.x, 0.75)
))) %>%
 pivot_longer(everything(), names_to = c("variable", "cuartil"), names_sep = "_") %>%
 pivot_wider(names_from = cuartil, values_from = value)
```

Warning: Expected 2 pieces. Additional pieces discarded in 12 rows [4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15].

Warning: Values from `value` are not uniquely identified; output will contain list-cols.

- \* Use `values\_fn = list` to suppress this warning.
- \* Use `values\_fn = {summary\_fun}` to summarise duplicates.
- \* Use the following dplyr code to identify duplicates.

```
{data} |>
 dplyr::summarise(n = dplyr::n(), .by = c(variable, cuartil)) |>
 dplyr::filter(n > 1L)
```

```
A tibble: 5 x 7
 variable Q1 Q2 Q3 hours rate score
 <chr> <list> <list> <list> <list> <list> <list>
1 age <dbl [1]> <dbl [1]> <dbl [1]> <NULL> <NULL> <NULL>
2 study <NULL> <NULL> <NULL> <dbl [3]> <NULL> <NULL>
3 attendance <NULL> <NULL> <NULL> <NULL> <dbl [3]> <NULL>
4 previous <NULL> <NULL> <NULL> <NULL> <NULL> <dbl [3]>
5 final <NULL> <NULL> <NULL> <NULL> <NULL> <dbl [3]>
```

### 3.3 Deciles y percentiles de final\_score

```
quantile(df$final_score, probs = seq(0.1, 1, by = 0.1))
```

```
10% 20% 30% 40% 50% 60% 70% 80% 90% 100%
36.9 43.0 47.0 52.0 56.0 61.0 65.0 71.0 76.0 95.0
```

```
quantile(df$final_score, probs = c(0.25, 0.50, 0.75, 0.90, 0.95))
```

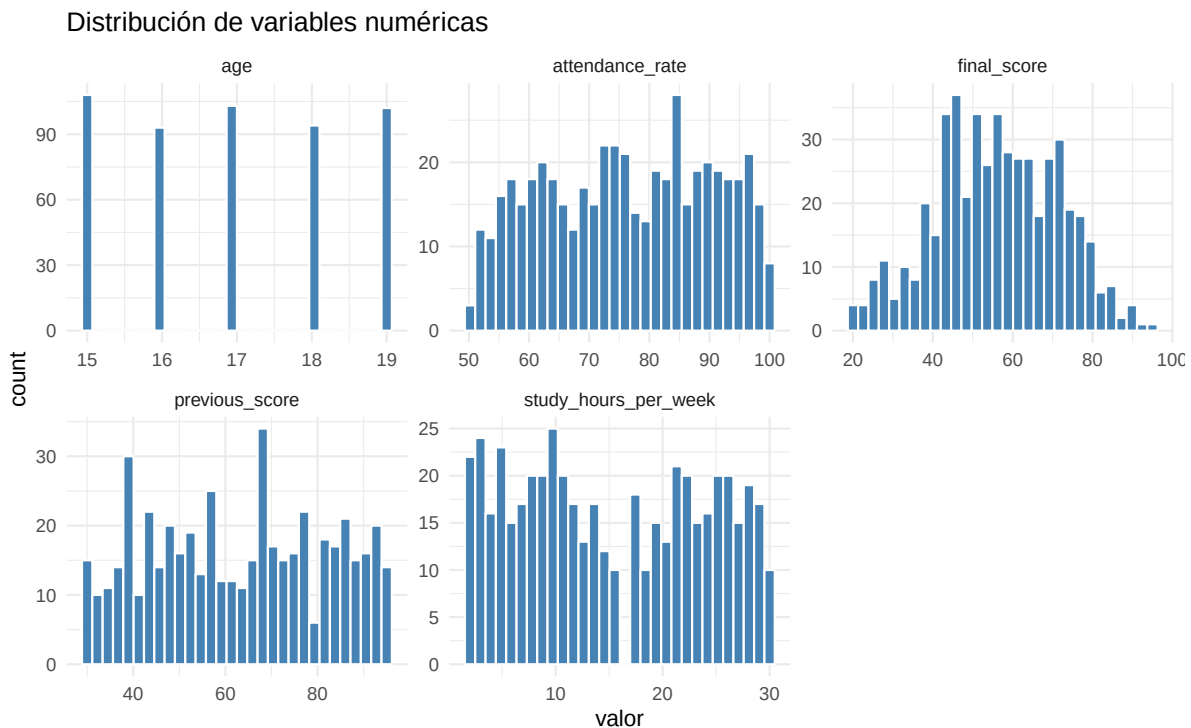
25% 50% 75% 90% 95%  
 45 56 68 76 80

#### 4. Histogramas y boxplots Histogramas

```
vars_num <- df %>% select(age, study_hours_per_week, attendance_rate, previous_score, final_score)

p1 <- vars_num %>%
 pivot_longer(everything(), names_to = "variable", values_to = "valor") %>%
 ggplot(aes(x = valor)) +
 geom_histogram(fill = "steelblue", color = "white", bins = 30) +
 facet_wrap(~ variable, scales = "free") +
 labs(title = "Distribución de variables numéricas") +
 theme_minimal()

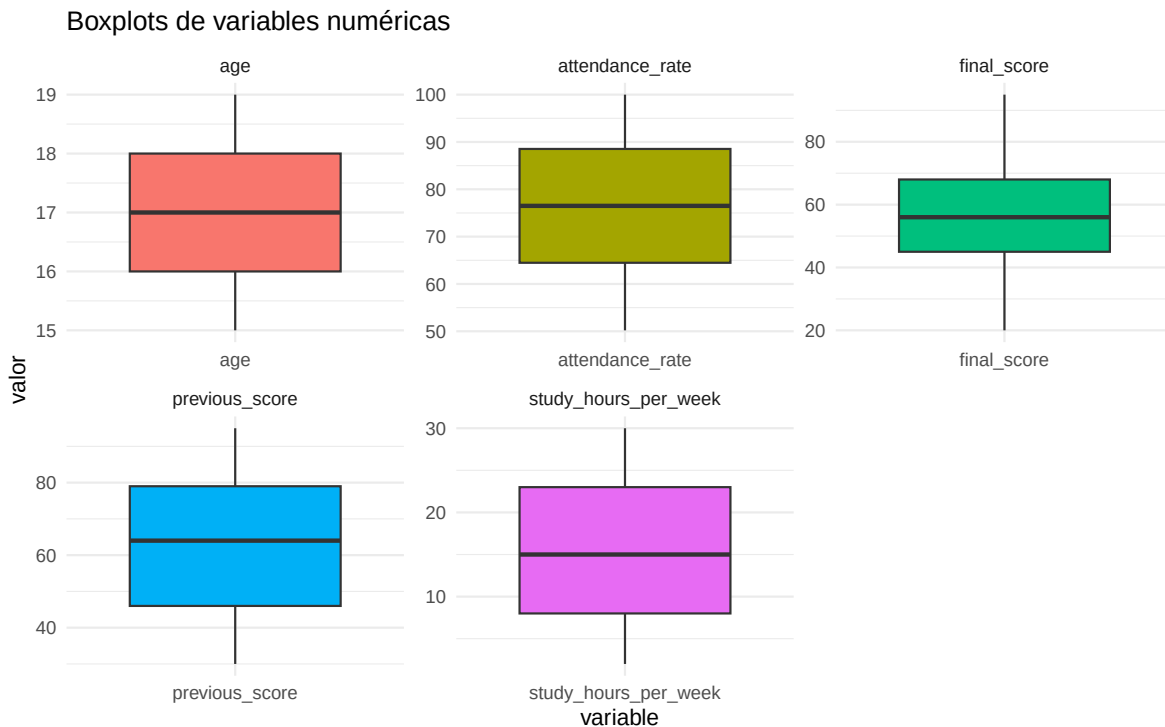
print(p1)
```



#### Boxplots

```
p2 <- vars_num %>%
 pivot_longer(everything(), names_to = "variable", values_to = "valor") %>%
 ggplot(aes(x = variable, y = valor, fill = variable)) +
 geom_boxplot() +
 facet_wrap(~ variable, scales = "free") +
 labs(title = "Boxplots de variables numéricas") +
 theme_minimal() +
 theme(legend.position = "none")

print(p2)
```



## 5. Análisis detallado: study\_hours\_per\_week 5.1 Tabla de frecuencias

```
breaks <- seq(min(df$study_hours_per_week), max(df$study_hours_per_week), length.out = 9)

amplitud <- breaks[2] - breaks[1]

tabla_freq <- df %>%
 mutate(intervalo = cut(study_hours_per_week, breaks = breaks, include.lowest = TRUE)) %>%
 group_by(intervalo) %>%
```

```

summarise(frec_abs = n(), .groups = "drop") %>%
mutate(
 frec_rel = frec_abs / sum(frec_abs),
 frec_porc = frec_rel * 100,
 frec_acum = cumsum(frec_abs)
)

tabla_freq

```

```

A tibble: 8 x 5
 intervalo frec_abs frec_rel frec_porc frec_acum
 <fct> <int> <dbl> <dbl> <int>
1 [2,5.5] 85 0.17 17 85
2 (5.5,9] 72 0.144 14.4 157
3 (9,12.5] 62 0.124 12.4 219
4 (12.5,16] 52 0.104 10.4 271
5 (16,19.5] 43 0.086 8.6 314
6 (19.5,23] 69 0.138 13.8 383
7 (23,26.5] 56 0.112 11.2 439
8 (26.5,30] 61 0.122 12.2 500

```

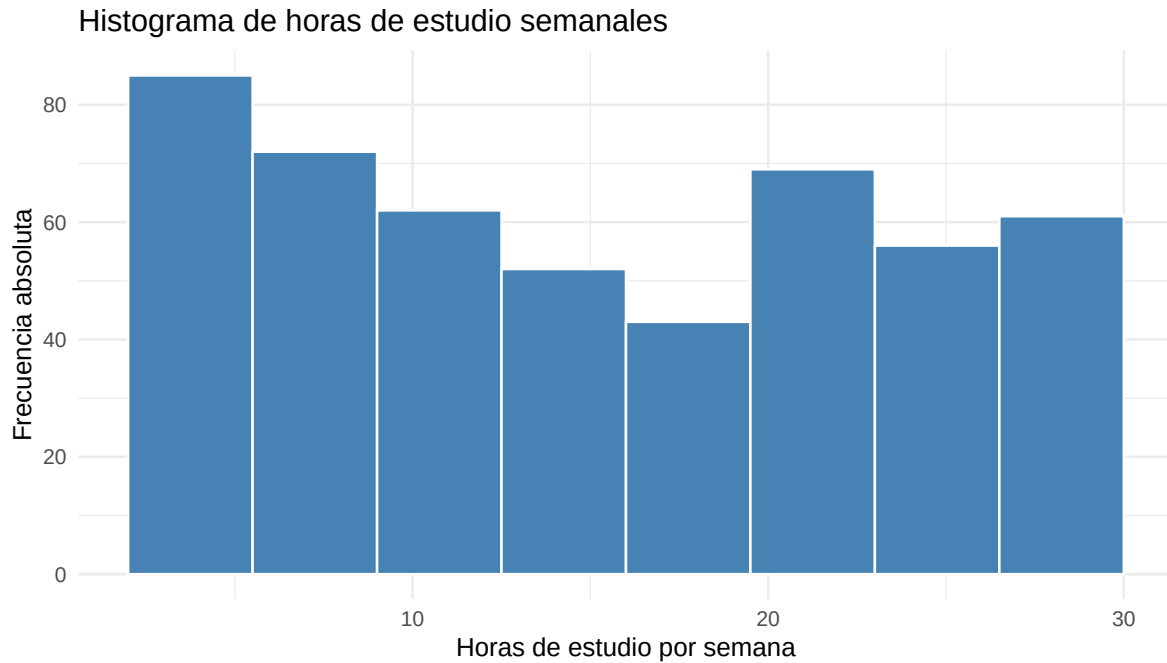
## 5.2 Histograma con intervalos de clase

```

p3 <- ggplot(df, aes(x = study_hours_per_week)) +
 geom_histogram(breaks = breaks, fill = "steelblue", color = "white") +
 labs(title = "Histograma de horas de estudio semanales",
 x = "Horas de estudio por semana",
 y = "Frecuencia absoluta") +
 theme_minimal()

print(p3)

```



### 5.3 Interpretación

La variable `study_hours_per_week` tiene una amplitud de clase de `r round(amplitud, 2)` horas. La distribución muestra que la mayoría de los estudiantes estudia entre 5 y 20 horas semanales.

El histograma permite detectar si hay asimetría en los datos: si la cola es más larga hacia la derecha hay sesgo positivo, y si es hacia la izquierda hay sesgo negativo. El valor de asimetría para esta variable es `r round(skewness_val, 2)`.